

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23532- COMPUTER NETWORKS LABORATORY

LAB MANUAL

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Practical - 1

Aim:
Study of various network commands used in Linux and Windows.

Basic Networking Commands:

① arp -a: ARP is short form of address resolution protocol, It will show the IP address of your computer along with the IP and MAC address of your router.

o/p
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

② hostname: This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

o/p
fedora

W ③ ipconfig /all: This command displays detailed configuration information about your TCP/IP connection including Router,

Gateway, DNS, DHCP, and type of ethernet adapter in your system.

o/p

```
Host Name . . . . . : DESKTOP-RKBICRNH
Physical Address . . . . . : 0A-00-27-00-00-06
DHCP Enabled . . . . . : No
IPv4 Address . . . . . : 192.168.56.1 (Preffered)
Subnet Mask . . . . . : 255.255.255.0
```

④ nbtstat -a : This command helps solve problems with NetBIOS name resolution.

o/p

```
NBTSTAT [[-a RemoteName] [-A IP address] [-c]
        [-n] [-r] [-R] [-RR] [-s] [-S]
        [interval]]
```

⑤ netstat : netstat displays a variety of statistics about a computer's active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics.

o/p

```
Active Internet Connections (w/o servers)
Proto  Recv-Q  Send-Q  Local address  Foreign address
```

⑥ nslookup: It is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. ns lookup can operate in two modes: interactive and non interactive

eg: nslookup www.google.com

o/p

Server : 127.0.0.53
Address : 127.0.0.53 #53

W ⑦ pathping: Pathping is unique to windows and is basically a combination of the ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

o/p

pathping [-g host-list] [-h maximum_hops] [-i address]
[-n] [-p period] [-q num_queries]
[-w timeout] [-4] [-6] target_name

⑧ ping: (Packet INternet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to other devices.

1. # ping hostname
2. # ping ip address
3. # ping fully qualified domain name.

ping fedora

o/p

PING fedora (fe80::aef9:32ef:d5e3:coc%.enp1s3)
icmp_seq=1 ttl=64 time=0.102 ms
icmp_seq=2 ttl=64 time=0.119 ms

⑨ Route: route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

o/p

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use
default	_gateway	0.0.0.0	UG	100	0	0
10.0.2.0	0.0.0.0	256.255.255.0	U	100	0	0

Some Important Linux networking commands:

1. ip:

ip <options> <object> <commard>

a) To show the ip address assigned to an interface on your server:

ip address show

b) To assign an ip to an interface.

ip address add 192.168.1.254/24 dev enp803

c) To delete an ip on an interface

ip address del 192.168.254/24 dev enp803

d) Alter the status of the interface by bringing the interface eth0.

ip link set eth0 up

e) Alter the status of the interface by bringing the interface eth0.

ip link set eth0 down

f) Alter the status of the interface by enabling promiscuous mode for eth0:

ip link set eth0 promisc on

g) Add a default route via the local gateway 192.168.1.254 that can be reached on device eth0:

ip route add default via 192.168.1.254 dev eth

h) Add a route to 192.168.1.0/24 via the gateway 192.168.1.254:

ip route add 192.168.1.0/24 via 192.168.1.254

i) Add a route to 192.168.1.0/24 that can be reached on device eth0:

ip route add 192.168.1.0/24 dev eth0.

j) Delete the route for 192.168.1.0/24 via the gateway at 192.168.1.254:

ip route delete 192.168.1.0/24 via 192.168.1.254

k) Display the route taken for ip 10.10.1.4:

ip route get 10.10.1.4

Outputs:

a) enp0s3:
inet 10.0.2.15/24 brd 10.0.2.255 scope global
dynamic noprefixroute enp0s3

inet6 fd17:625c:f037:2:7f31:2eb6:74e3:d744/64

inet 6 fe80::aef9:32ef:d5e3:c0c/64

b) enp0s3:
inet 10.0.2.15/24 brd 10.0.2.255

inet 192.168.1.254/24

inet 6 fd17:625c:f037:2:7f31:2eb6:74e3:d744/64

inet 6 fe80::aef9:32ef:d5e3:c0c/64

d) enp0s3: <Broadcast, Multicast, UP, Lower_Up> mtu 1500
qdisc fq_codel state UP mode default group
qlen 1000
link/ether 08:00:27:2d:d7:3c brd ff:ff:ff:ff:ff:ff
altname enx080027 2dd73c

e) enp0s3: <Broadcast, Multicast> mtu 1500 qdisc fq_codel
state DOWN mod Default group default qlen 1000

f) enp0s3: <Broadcast, Multicast, PROMISC> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000

g) enp0s3: <Broadcast, Multicast, Promisc, UP, Lower_up> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:2d:d7:3c brd ff:ff:ff:ff:ff:ff
altname enx080027272dd73c
inet 192.168.1.254/24 scope global enp0s3
valid_lft forever preferred_lft forever

h) RTNETLINK answers: File exists

i) RTNETLINK answers: File exists.

K) 10.10.1.4 dev enp0s3 src 192.168.1.254 uid 0
cache

2. ifconfig:

The ifconfig is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has been replaced by the ip command discussed above.
o/p

enp0s3: flags=4419<UP, Broadcast, Running, Multicast> mtu 1500
inet 192.168.1.254 netmask 255.255.255.0
ether 08:00:27:2d:d7:3c txqueuelen 1000

3. mtr:

* MTR combines ping and traceroute into a single command-line network diagnostic tool.
It maps the network path to a destination host, displaying real-time statistics like response time and packet loss for each hop.

mtr <options> hostname/IP

a) mtr google.com
The basic mtr command shows you the statistics, including each hop with time and loss%.

o/p

	Host	Loss%	Snt	Last	Avg	Best	W
1.	gateway.local	0.0%.	10	0.6	0.7	0.5	1
2.	10.10.0.1	0.0%.	10	2.1	2.3	2.0	3
3.	isp-router.example.net	0.0%.	10	8.5	8.9	8.2	1
4.	core1.isp.net	0.0%.	10	15.4	15.8	15.0	1
5.	google-peer.isp.net	0.0%.	10	21.7	22.1	21.3	2

(b) Show numeric ip addresses instead of host names.
mtr -g google.com

o/p

1.	192.168.1.1	0.0%	10	0.6	0.7	0.5
2.	10.10.0.1	0.0%	10	2.1	2.3	2.0
3.	203.0.113.1	0.0%	10	8.5	8.9	8.2

c) Shows IP addresses and host names.

mtr -b google.com

o/p

1.	gateway.local (192.168.1.1)	0.0%	10	0.6	0.7	
2.	10.10.0.1 (10.10.0.1)	0.0%	10	2.1	2.3	
3.	isp-router.example.net (203.0.113.1)	0.0%	10	8.5	8.9	

d) Send the number of pings you want to send.

mtr -c 10 google.com

o/p

Start: 2026-07-28T12:00:00+0000

Host : server.example.com

4. tcpdump:

* The tcp command is designed for capturi and displaying packets

Installation :

dnf install -y tcpdump

* To know which interfaces tcp dump can use.

tcpdump -D

* To capture traffic of eth0, you can initiate that with tcp dump -i eth0

tcpdump -i eth0

o/p

12:15:01:123456 IP 192.168.1.100.52344 > 8.8.8.8.5
:12345+ A? google.com. (28)
12:15:01.12[redacted]5678 IP 8.8.8.8.53 > 192.168.1.100:12345
1/0/0 A 142.250.193.14 (44)

a) Capture 10 packets and exit

tcpdump -i eth0 -c 10

o/p

tcpdump : listening on eth0, link-type EN
12:30:01.123456 IP 192.168.1.10.54321 > 8.8.8.53:DN
12:30:01:125678 IP 8.8.8.[redacted].53 > 192.168.1.10.54321:
⋮
10 packets captured

b) Traffic to and from host 8.8.8.8

tcpdump -i eth0 -c 10 host 8.8.8.8

o/p

12:35:01.123456 IP 192.168.1.10.53210 > 8.8.8.8.53:

⋮

12:35:05.346890 IP 8.8.8.8 > 192.168.1.10: ICMP echo

c) Traffic to and from ~~host~~ network 10.1.0.0/24

tcpdump -i eth0 net 10.1.0.0/24

o/p

12:45:01.123456 IP 10.1.0.25.443 > 192.168.1.10.55000: H

d) Traffic to and from port numbers.

tcpdump -i eth0 port 53

o/p

12:50:01.123456 IP 192.168.1.10.5400 > 8.8.8.8.53: D

5. Ping:

* Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control Message Protocol Echo Request messages.

* Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution.

```
# ping google.com

PING google.com (216.58.206.174) 56(84) bytes
of data.
64 bytes from sof02s27-in-f14.1e100.net (216.58.20
icmp-seq=1 ttl=56 time=10.7 ms
:
:
```

* To stop press ctrl + C.

* To ping a host 10 times,

```
ping -c 10 google.com
```

→ Configuring an Ethernet connection by using nmcli:

* If you connect a host to the network o Ethernet, you can manage the connect

settings on the command line by using the nmcli utility.

Procedure:

1. List the Network Manager connection profiles:

nmcli connection show

o/p

Name	UUID	Type	Device
Wired 1	a5eb6490-cc20-3668-81	ethernet	enp1s0

2. Display the current settings of the connection.

#nmcli connection show

o/p

connection.interface-name: enp1s0
connection.autoconnect: yes
ipv4.method: auto
ipv6.method: auto.

3. Configure the ipv4 settings

* To use DHCP enter,
nmcli connection modify "Wired 1" ipv4.method aut

* To set a static IPv4 address, network mask, default gateway, DNS server,

and ~~sear~~ search domain enter,

nmcli connection modify "Wired 1" ipv4.method manual ipv4 addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com.

6. Configure the IPV6 settings:

* To set a static IPV6 address, network mask, default gateway, DNS servers and search domain enter:

nmcli connection modify "Wired 1" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::fffe ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com.

7. Activate the profile:

nmcli connection up Internal_LAN

Verification

(1) Display the IP settings of the NIC:

ip address show enp1s0

o/p
enp1s0 : <Broadcast, Multicast, UP, Lower_UP>
mtu 1500 qdisc

② Display the IPV4 default

ip route show default

o/p
default via 192.0.2.254 dev enp1s0 proto
static metric 102

③ Display the IPV6 default

ip -6 route show default

o/p
default via 2001:db8:1::ffee dev enp1s0
proto static metric 102 pref medium

④ Display the DNS settings

cat /etc/resolv.conf

o/p
search example.com
nameserver 192.0.2.200
nameserver 2001:db8:1::ff66

⑤ Use the ping utility to verify that this host can send packets to other hosts.

ping <host-name-or-IP-address>

Student Observation:

① Which command is used to find the reachability of a host machine from your device?

ping: example: ping google.com

② Which command gives the details of hops taken by a packet to reach its destination?

traceroute: example: traceroute google.com

③ Which command displays the IP config of your machine?

ip addr show
(or)
ifconfig

④ Which command displays the TCP port status in your machine?

⇒ netstat command.

⑤ Write the command to modify the IP config in a linux machine.

sudo ip addr add 192.168.1.100/24 dev eth0

Maximum | Advantages | Applications |

eth0

Practical - 2

Aim: Study of different types of network cable

a) Understand different types of network cable.

1. Unshielded twisted pair (UTP) cable
2. Shielded twisted pair (STP) cable
3. Coaxial cable
4. Fibre optic cable

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Applications/ Use	Image
UTP	Category 3	10 bps	Advantages * Cheaper in cost * Easy to install as they have a small overall diameter. Disadvantages * More prone to Electromagnetic interference & noise	10 Base -T Ethernet	
	Category 5	Upto 100 Mbps		Fast ethernet, Gigabit ethernet	
	Category 5e	1 Gbps		Fast ethernet, Gigabit ethernet	
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded! * Faster than UTP * Less susceptible to noise & interference Disadvantage: * Expensive * Greater installation effort	Gigabit ethernet, 10 G ethernet (55m) Widely used in data centers	
SSTP	Category 7	10 Gbps		Gigabit ethernet, 10 G Ethernet (100 m)	

Coaxial cable	RG-6, RG-59, RG-11	10 - 100 Mbps	* High Bandwidth * Immune to interference. * Low loss Bandwidth * Versatile Disadvantage: * Limited distance * Cost * Size is bulky	Speed of signal is 500 m. Television network. High speed internet connections.
Fibre Optics Cable	Single mode Multi mode	100 Gbps	Advantages: * High speed * High bandwidth * High security * Long distance	* Maximum distance of fibre optics cable is around 100 meters.

b) Make Your Own Ethernet Cross-Over Cable / Straight cable

Tools and parts needed:

* Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.

* A crimping tool. This is an all in one networking tool shaped to push down the pins in the plug and strip and cut the shielding of the cables.

* Two RJ45 plugs

* Optional two plug shields.

Step 1: To start construction of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheets, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away.

Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding.

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeat for the other end diagram B / A.

To test it, plug it in and attempt to connect two devices directly.

Practical - 3

Aim: To study the Packet tracer tool Installation and User Interface Overview.

To understand environment of CISCO Packet Tracer to design simple network.

Introduction:

A simulator, as the name suggests, simulates network devices and its environment. Packet Tracer is an exciting network design, simulation and modelling tool.

1. It allows you to model complex systems without the need for dedicated equipment.

2. It helps you to practice your network configuration and troubleshooting skills via computer or an android or IOS based mobile device.

3. It is available for both the Linux and windows desktop environments.

4. Protocols in Packet tracer are coded to work and behave in the same way as they would on real hardware.

User interface overview :

The layout of packet tracer interface is divided into several components. The components of the packet tracer interface are as follows :

1) Menu Bar - This is a common menu found in all software applications ; it is used to open, save, print, change preferences, and so on.

2) Main Toolbar - This is a common menu found in all software applications ; it is used to open, save, print, change preference, and so on.

3) Logical / Physical workspace tabs - These tabs are ~~used~~ allow you to toggle between the Logical and Physical work areas.

4) Workspace - This is the area where topologies are created and simulations are displayed.

5) Common tools bar - This toolbar provides controls for manipulating topologies, such as select, move layout, place note, delete, inspect, resize shape.

6) Real time / Simulation: These tabs are used to toggle between the real and simulation modes. Buttons are also provided to control the time, and to capture the packets.

7) Network component box: This component contains all of the network and end devices available with Packet tracer, and is further divided into two areas
Area 7a: Device type selection box - This area contains device categories
Area 7b: Device specific selection box - When a device category is selected, this selection box displays the different device models within that category.

8) User-created packet box - Users can create highly customised packets to test their topology from this area, and the results are displayed as a list.

→ Analyse the Behaviour of network device using CISCO packet tracer simulation.

1. From the network component box, click and drag-and-drop the below components:
   a. 4 generic PCs and one hub
   b. 4 generic PCs and one switch.

2. Click on connections:
a. Click on Copper Straight - Through cable,
b. Select one of the PC ~~or~~ and connect it to the hub using the cable. The link LED should glow in green, indicating that the link is up. Similarly connect remaining 3 PCs to the ~~##~~ hub.
c. Similarly connect 4 PCs ~~#~~ to the switch using copper straight - Through cable.

3. Click on PCs connected to hub, go to Desktop tab, click on IP configuration, and enter an IP address and subnet mask. Here, the default gateway and DNS server information is not needed as there are only two ~~ends~~ devices in the network

Click on the PDU from the common tool bar,
a. Drag and drop it on one of PC and then drop it on another PC connected to the Hub.

4. Observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation.

5. Repeat step #3 to step #5 for the PCs connected to the switch.

6. Observe how hub and switch are forwarding the PDU and write your

observation and conclusion about the behaviours of switch and hub.

1.

2.

1)

2)

3)

3.

Practical - 4

Setting up a Local Area Network (LAN)

1. **Aim:**
To design, setup & configure a local Area Network connecting four ~~cop~~ computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network.

2. **Equipment Required:**

1)	Personal computers	4	Ethernet (NIC) enabled
2)	Network Switch	1	8/16/24 port, unmanaged /managed.
3)	Ethernet Cables (cat 5e / Cat 6)	4 8	Straight - through / RJ-45 connectors.

3. **Theory:**

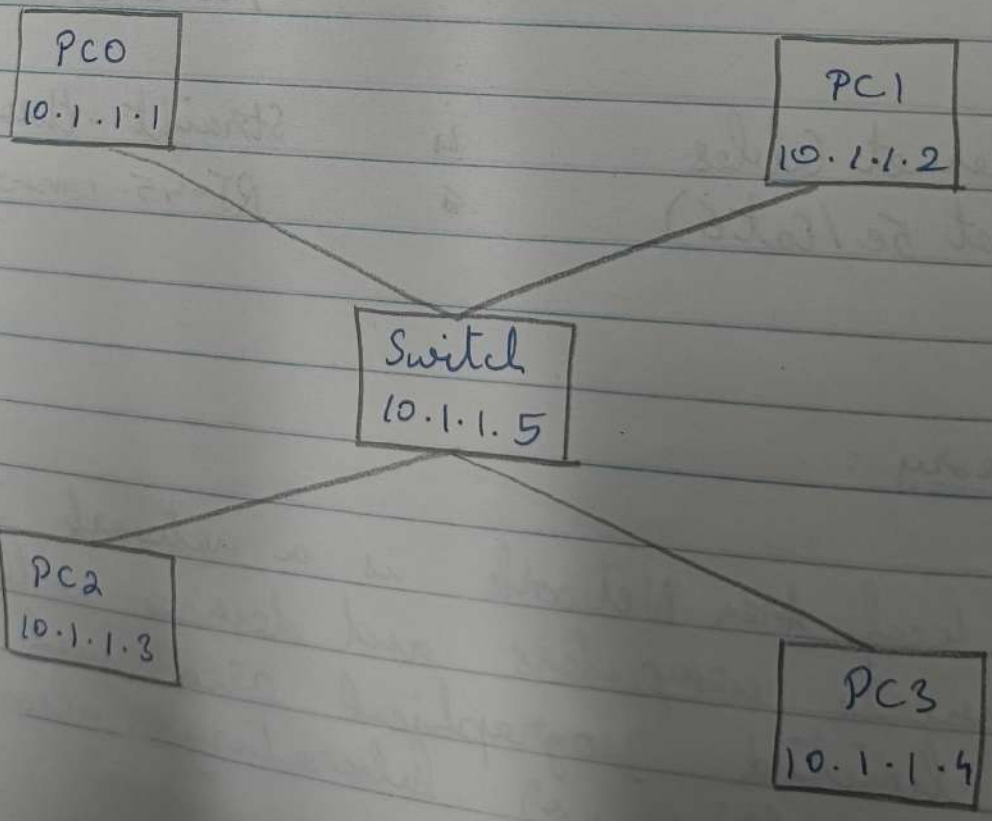
* A local Area Network is a network that connects computers and devices within a limited geographical area such as home, office, or laboratory, allowing

them to share resources.

A star topology is used in this experiment where all end devices are individually connected to a central network switch. The switch operates at Layer 2 of the OSI model and forwards data frames between connected devices based on their MAC addresses. Every device on a LAN must be assigned an IP address within the same network range.

4. Network Diagram

The LAN was implemented using a Star Topology as shown.



5. IP Address Configuration Table :

Device	IP Address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 interface

6. Implementation in Cisco Packet Tracker :

Step 1: Place the devices

a) Open Cisco Packet Tracker
b) From the device category panel, select end devices and drag 4 PCs onto the workspace.
c) Select switches and drag one switch.

Step 2: Cable the Devices

d) Click connections → select Copper Straight through
e) Click each PC, then click switch, and select available FastEthernet port to complete each connection.

Step 3: Assign IP addresses to PCs

f) Click a PC → desktop tab → IP configuration
g) Enter the static IP address and subnet mask.

Step 4: Configure the Switch Management IP

h) Click the switch → CLI tab
i) Enter the following commands
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit

Step 5: Verify Connectivity Using Ping

j) Click a PC → Desktop tab → Command Prompt.
k) Use the ping command to test connectivity to the other PCs eg,
ping 10.1.1.2.
l) Successful "Reply from..." messages confirm connectivity.

Step 6: Demonstrate Data Sharing (Server Simulation)

m) Add a server device (End Devices → Server-PT) and connect it to the switch.
n) Assign it an IP address (eg; 10.1.1.10, subnet mask 255.0.0.0) via its Desktop → IP Config tab.
o) Open the server's services tab and enable the HTTP service.

Step 7: Observe Packet Flow in Simulation mode

p) Switch to simulation mode.
2) Send a ping or web request between two devices and use the Play / Capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

7. Observation:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet (10.0.0.0 / 255.0.0.0). Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder / server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.

Practical - 6

Aim: Experiments on Packet Capture

Tool: Wireshark

Packet Sniffer:

* Sniffs messages being sent / recieved from / by your computer
* Store and display the contents of the various protocol fields in the messages
* Passive programs
  - never sends packets itself
  - no packets addressed to it
  - recieves a copy of all packets

Packet Sniffer Diagnostic tools:

* Tcpdump
  - eg. tcpdump -enx host 10.129.41.2 -w exe.out
* Wireshark
  - wireshark -r exe3.out

Description:

* Wireshark, a network analysis tool formerly known as ethereal, captures packets in realtime and display them in human readable format. Wireshark includes filters, colour coding, and other features that lets you dig deep into network traffic and inspect individual packets.

What we can do with wireshark:

* Capture network traffic
* Decode packet protocols using dissectors
* Define filters - capture and display
* Watch smart statistics.
* Analyze problems
* Interactively browse that traffic.

Capturing Packets and Analysing using wireshark tool:

To filter, capture, view, packets in wireshark tool.
Capture 100 packets from the ethernet: IEEE 802.3 LAN Interface and save it.

Procedure:

* Select Local Area Connection in Wireshark
* Go to capture option
* Select stop capture automatically after 100 packets
* Then click start capture.

Output

1. Create a filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

Procedure:

* Select Local Area Connection in wireshark
* Go to capture option
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Search TCP packets in search bar.
* To see flow graph click statistics flow graph.

Flow Graphs:

2. Create a filter to display only ARP packets and inspect the packets.

Procedure:

* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Searel ARP packets in search bar.
* Save the packets.

Output:

3. Create a filter to display only DNS packets and provide the flow graph.

Procedure:

* Go to capture option
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Search DNS packets in search bar.
* To see flow graphs click statistics flow graph.
* Save packets.

4. Create a filter to display only HTTP packets and inspect the packets procedure.

Procedure:

* Selet local Area Connection in wireshark
* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture
* Search HTTP packets in search bar.

5. Create a filter to display only IP/ICMP packets and inspect the packets.

Procedure:

* Select Local Area # Connection in wireshark
* Go to capture option
* Select stop capture automatically after 100 packets.
* Then click start capture.
* ~~Send~~ Seared ICMP/IP packets in search bar.
* Save Packets.

6. Create a filter to display only DHCP packets and inspect the packets.

Procedure:

* Select local Area connection in wireshark
* Go to capture option
* Select stop capture automatically after 100 packets.
* Then click start capture
* Search DHCP packets in search bar
* Save the packets.

Practical - 6

Aim: Write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

Error Correction at Data Link Layer:

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the reciever. It is a technique developed by R. W. Hamming for error correction.

Create sender program with:

1. Input to sender file should be a text of any length. Program should convert the text to binary.
2. Apply hamming code concept on the binary data and add redundant bits to it.
3. Save this output in a file called channel.

Create a reciever program with :

1. Reciever program should read the input from channel file.
2. Apply hamming code on the binary data to check for errors.
3. If there is an error, display the position of the error. Else remove the redundant bits and convert the binary data to ascii and display.

Student Observation :

Code : Sender Program :

```
text = input (" Enter text : ")
data = " "
for ch in text:
    data += format (ord (ch), "08b")
m = len (data)
r = 0
while 2**r < m+r+1 :
    r += 1
n = m + r
code = [0] * (n+1)
j = 0
for i in range (1, n+1):
    if i & (i-1):
        code [i] = int (data [j])
        j += 1
```

```
for i in range(r):
    p = 2**i
    parity = 0
    for j in range(1, n+1):
        if j & p:
            parity ^= code[j]
    code[p] = parity
hamming = ""
for i in range(1, n+1):
    hamming += str(code[i])
with open("channel.txt", "w") as f:
    f.write(hamming)
print("Binary Data:", data)
print("Hamming Code:", hamming)
print("Data saved to channel.txt")
```

Code : Reciever Program :

```
with open("channel.txt", "r") as f:
    data = f.read().strip()
code = [0] + [int(x) for x in data]
n = len(data)
r = 0
while 2**r < n+1:
    r += 1
error = 0
for i in range(r):
    p = 2**i
    parity = 0
    for j in range(1, n+1):
        if j & p:
            parity ^= code[j]
```

```
            if parity:
                error += p
    if error != 0:
        print("Error detected at position :", error)
        code[error] ^= 1
        print("Error corrected.")
    else:
        print("No error detected.")
    result = ""
    for i in range(1, n+1):
        if i & (i-1):
            result += str(code[i])
    output = ""
    for i in range(0, len(result), 8):
        binary = result[i:i+8]
        output += chr(int(binary, 2))
    print("Output:", output)
```

Input:

```
Enter text: Hello
Binary data : 0100100001000101010011.....
Hamming code: 0000100100100000110100 1....
Data saved to channel.txt
```

Output:

```
No error detected
Output: Hello
```

Practical – 7

Aim: Write a program to implement flow cont at data link layer using Sliding Windo Protocol. Simulate the flow of frames fr one node to another.

Create a sender program with:

1. Input Window size from the user.
2. Input a text message from the user.
3. Consider 1 character per frame.
4. Create a frame with fields
5. Send the frames
6. Wait for aknowledgement from the reciever.
7. Reader a file called Receivers_buffer.
8. Check ACK field for Aknowledgemant number.

Create a recievers file with:

1. Reader a file called Sender_Buffer.
2. Check the frame no.
3. If the Fame no. are as expected, write the appropriate ACK no. in the Reciever_Buffer fo Else write NACK no. in the Reciever_Buffer f

NOTE: Induce error and verify the behaviour o the program. Manually change the frame no and ACK no in the files.

Student Observation :

Sender Program Code :

```
import time
msg = input ("Enter the text :")
w = int (input ("Enter window size :")
frames = []
for i in range (len (msg)):
    frames.append ([i+1, msg [i]])
start = 0
while start < len (frames):
    end = min (start + w, len (frames))
    F = open ("Sender_Buffer.txt", "w")
    print ("\n Frames sent :")
    for i in range (start, end):
        print ("Frame no :", frames [i][0],
               "DATA :", frames [i][1])
        f.write (str (frames [i][0]+ "" + frames [i][1])
    f.close()
    print ("Waiting for aknowledgement...")
    time.sleep (3)
    try:
        f = open ("Reciever_Buff.txt", "r")
        ack = int (f.read().strip ())
        f.close()
    except:
        print ("No ACK recieved")
        continue
print ("Recieved ACK :", ack)
```

```
        if ack == end:
            print ("ACK correct. Sending next window.")
            start = end
        else:
            print ("NACK recieved.")
            print ("Resending Frames...")
            time.sleep(2)
    print ("\n All frames transmitted")
    print ("Message :", msg)
```

Recieves Program Code:

```
import time
expected = 1
while True:
    try:
        f = open ("Sender_Buffer.txt", "r")
        lines = f.readlines()
        f.close()
    except:
        print ("Sender buffer not found")
        time.sleep(2)
        continue
    if len(lines) == 0:
        continue

error = False
print ("\n Frames recieved:")
```

```
        for line in lines:
            parts = line.split()
            data = parts[1]
            print("Frame no:", frame, "DATA:", data)
            if frame != expected:
                error = True
                break
            expected += 1
    f = open("Reciever_Buffer.txt", "w")
    if error:
        print("Wrong Frame recieved.")
        print("Sending NACK:", expected)
        f.write(str(expected))
    else:
        print("All frames recieved.")
        print("Sending ACK:", expected)
        f.write(str(expected))
    f.close()
    time.sleep(2)
    if expected > 1000:
        break.
```

Expected Output:

```
Enter the text: HELLO
Enter window size: 3
Frames sent:
Frame no: 1 DATA: H
Frame no: 2 DATA: E
Frame no: 3 DATA: L
```

Waiting for aknowledgement....
Recieved ACK: 4
ACK correct. Sending next window.

Frames sent:
Frame no: 4 DATA: L
Frame no: 5 DATA: O

Waiting for aknowledgement....
Recieved ACK: 6
ACK correct. Sending next window.

All frames Transmitted.
Message: HELLO

Practical -8

Nmap Live Host Discovery

Aim: To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

Introduction:

Nmap is a free, open-source network-scanning tool created by Gordon Lyon and first released in 1997. It is the industry standard tool for mapping networks, discovering live hosts, and identifying the services running on them.

Learning Objectives:

* Understand why host discovery must precede port scanning.
* Learn how ARP scans work at the link layer using broadcast requests.
* Understand how ICMP echo, timestamp, and address-mask requests reveal live hosts across subnets.

* Learn how TCP SYN and ACK ping scans works, and the difference between privileged and unprivileged scanning behaviour.

* Understand how UDP ping scans use ICMP port-unreachable responses to confirm a host is online.

Prerequisites:

* A tryhackme account with access to the "Nmap Live Host Discovery" room.

* Basic familiarity with the TCP/IP model and IP subnetting.

* A basic linux terminal.

Step-by-step Lab Checklist:

1. Log into TryHackMe and open the "Nmap Live Host Discovery" room.

2. Work through tasks 1-3 and answer the simulator based questions.

3. Deploy the lab network in task 4 and start the attack box; note the target subnet/ CONNECTION_IP.

4. Run an ARP scan: sudo nmap -PR -sn <subnet>/24 and record how many hosts respond.

5. Run an ICMP echo scan: sudo nmap -PE -sn <subnet>/24 compare the result with the ARP scan.

6. Run ICMP Timestamp (-PP) and ICMP address-Mask (-PM) scans; note which one are filtered.

7. Run a TCP SYN ping (-PS), TCP ACK ping (-PA, needs sudo), and a ~~##~~ UDP ping (-PU); compare results and required privilege level for each.

8. Optionally capture traffic with tcpdump or Wireshark on the tun() interface while each scan runs, to see the raw ARP/ICMP/TCP/UDP packets.

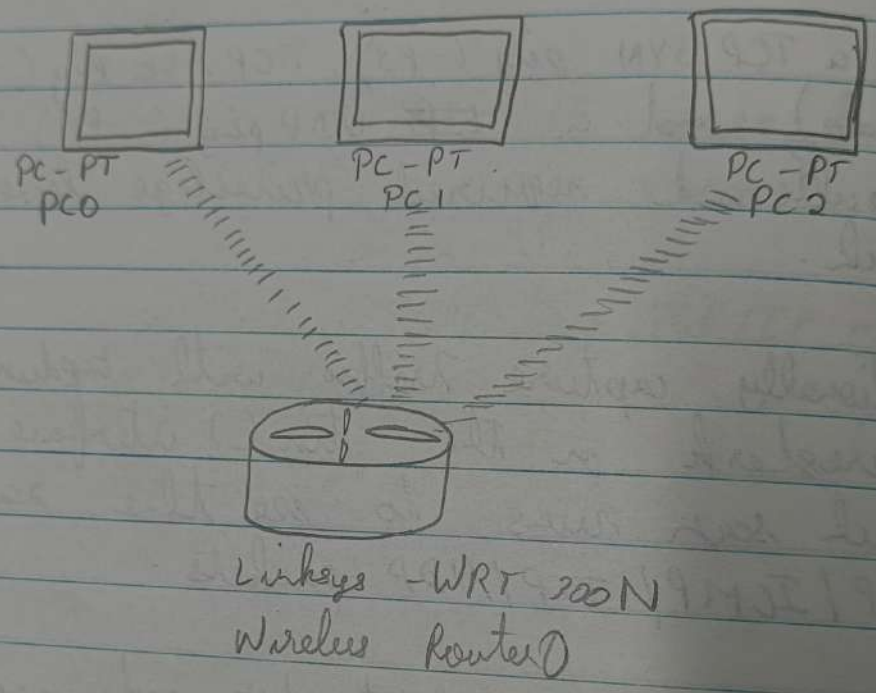
9. Try nmap -sL <subnet>/24 and nmap -R <subnet>/24 to see target listing and reverse-DNS behaviour.

10. Complete the room's knowledge-check questions in Tasks-5-9, then move on to the "Nmap Basic Port Scans" room.

Practical -8 (b)

Aim: Configuration of wireless LAN using Cisco Packet Tracer.

Design a topology with three PCs connected from Linksys Wireless Routers.



Perform following configuration:

* Configure stetic IP on PC and wireless Router.
* Set SSID to MotherNetwork
* Set IP address of router to 192.168.0.1, PC0 to 192.168.0.2, PC1 to 192.1.168.0.3 PC2 to 192.168.0.4.
* Secure your network by configuring WAP key on router.

* Connect Pc using WAP key on router.

To complete these tasks follow these step-by-step:

Step 1: Click on wireless router,

* Select Administration tab from top menu, set username and password to admin and click on save.

* Next click on wireless tab and set default SSID to Mother Network.

* Now select wireless security and change Security Mode to WEP.

* Set Key1 to 0123456789

* Again go in the end of page and click on save.

* Now we have completed all given tasks on wireless router. Now configure static IP on all three PCs.

* Double click on Pc select desktop tab click on IP config select static IP and set IP as given below.

* Now its time to connect PC's from wireless router. To do so click PC select desktop click on PC wireless.

* Click on connect tab and click refresh button.

* It will ask for WAP key insert 01234567 and click connect.

* It will connect you with wireless router

* Repeat same process on PC1 and PC2.

Practical -9

Aim: Implementation of subnetting in CISCO Packet tracer simulator.

Creating a Network Topology:

The first step in implementing classless IP subnetting is to create a network topology in packet tracer. To create a network topology in packet tracer, select the "New" button in the top left corner, then select "Network and "Generic".

Adding the devices:

Once we have created our network topology, we can add devices to it. To add a device, select device from the bottom left and drag into the network topology.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch and the router, we can use a /27 subnet mask.

Configuring the Devices:

Right click on the router and select "C
This will open the command line inte
for the router. In the CLI, enter th
following:

```
#enable
#configure terminal
#interface FastEthernet 0/0
#ip address {IP address} {subnet mask}
#no shutdown
#exit
```

```
interface FastEthernet 0/1
ip address {IP address} {subnet mask}
no shutdown
exit
```

Next we will configure the switch to ope
in access mode on its two ports, whic
are connected to the two PCs.
Finally, we will configure the PCs. Rig
click on the switch and click on "C
enter the following commands:

```
enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit.
```

interface FastEthernet 0/2
switchport mode access
exit

To configure the GigabitEthernet interface on the router, you can follow these steps:

1. Right-click on the router and select CLI
2. Enter the following:

enable
configure terminal
interface GigabitEthernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit

Testing the network:

Now that our topology is configured, we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful then the network is functioning properly.

Practical - 10

a) Aim: Internetworking with routers in CLI. PACKET TRACER simulator.

Procedure:

Step - 1:

1. Select the router and Open CLI
2. Press ENTER to start configuring Router
3. Type enable to activate the priviledged mode.

Router Command Line Interface:

```
Router > enable
Router # config t
Enter configuration commands, one per line.
Router (config) # interface FastEthernet 0/0
Router (config) # ip address 192.168.10.1
255.255.255.0 Router (config-if) # no shutdown
```

Step - 2 (Configuring PCs):

1. Assign IP addresses to every PC in the netwo
2. Select the PC, go to the desktop and select IP configuration and assign an IP address, Default gateway, Subnet mask.

3. Assign the default gateway of PC0 as 192.168.10.1
4. Assign the default gateway of PC1 as 192.168.20.1

Step-3 (Connecting PCs with Router):

1. Connect FastEthernet0 port of PC0 with FastEthernet 0/0 port of router 1 using a copper straight-through cable.

2. Connect FastEthernet0 port of PC1 with FastEthernet 0/1 port of router 1 using a copper straight-through cable.

Router Config Table:

Device Name	IP address FastEthernet 0/0	Subnet Mask	IP address FastEthernet 0/1	Subnet Mask
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.25

Practical - 11

Aim : a) Simulate Static Routing configuration using CISCO Packet tracer.

Setting up a practice lab:

Create a packet tracer lab as shown or download pre created ~~app~~ lab and load it.

Use the following commands to create two static routes for network 30.0.0.0 the route will place the first route to the routing table.

ip route 30.0.0.0 255.0.0.0 20.0.0.2 10

ip route 30.0.0.0 255.0.0.0 30.0.0.2 1

Creating, adding, verifying static routes

* We only need to add routes for the networks that are not available on the router's interface. Network 30.0.0.0/8 and network 50.0.0.0/8 are not available on router we have to create and add routes only for these networks.

Router 0 Requirements:

* Create two routes for network 30.0.0.0/8 and configure the first route as the main route and the second route as a back

* Create two routes for the host 30.0.0.100/8 and configure the first route as the main route and the second route as a backup route.

* Create two routes for network 50.0.0.0/8 and configure the first route as the main route and the second route as a backup route.

* Verify the router adds only main routes to the table.

Router 0 configuration:

Router > enable
Router # configure
terminal
Router (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.1
" " " " " " 40.0.0.2 20
" " " " 30.0.0.100 255.255.255.255
40.0.0.2 1
" " " " 20.0.0.2 20
" " " " 50.0.0.0 255.0.0.0 40.0.0.2 10
" " " " " 20.0.0.2 20

" " # exit
" " show ip route static

Router 1 requirements :

* Create two routes for network 10.0.0.0/8 ; configure the first route as the main route and the second route as a backup route.

* Create two routes for network 40.0.0.0 and configure the first route as the main route and the second route as a backup route.

Router 1 configuration :

Router >enable
Router # configure
terminal
Router(config) # ip route 10.0.0.0 255.0.0.0 20.0.
" " " " " " " 50.0.
" " " " " 40.0.0.0 255.0.0.0 20.0.
" " " " " " " 30.0.
Router (config) # exit.
" # show ip route static.

Router 2 requirements :

* Create static routes for network 20.0 and network 30.0.0.0/8 and verify the adds both routes to the routing table

Router 2 Configuration:

Router > enable
Router # configure
terminal
Router (config) # ip route 10.0.0.0 255.0.0.0 40.0.0.1
" " " " " 30.0.0.0 255.0.0.0
50.0.0.2
Router (config) # exit
" # show ip route static.

Deleting a static Route:

To delete a static route,

* use the 'show ip route static' command to print all static routes.
* Note down the route you want to delete.
* Use the 'no ip route' to delete the route.

Practical - 12

Aim: a) Implement echo client server using TCP/UDP sockets.

Algorithm:

Server:

1. Create a socket.
2. Bind the socket to an IP address and port number
3. Listen for client connections (TCP).
4. Accept the client connection (TCP).
5. Recieve the message from the client.
6. Send the same message back to the client.
7. Close the socket.

Client:

1. Create a socket
2. Connect to the server (TCP).
3. Read a message from the user.
4. Send the message to the server.
5. Recieve the echoed message
6. Display the recieved message
7. Close the socket.

Input:

Hello

Output:

Client:

Enter message: Hello
Server: Hello

Server:

Client: Hello.

Result:

Thus the echo client server was implemented successfully using TCP sockets.

Practical - 12

Aim: (b) To implement a dat client-server using TCP sockets.

Algorithm:

Server:

1. Create a socket
2. Bind the socket to an IP address and port.
3. Listen and accept all client connections
4. Recieve messages from client.
5. Send replies to the client.
6. Close the connection.

Client:

1. Create socket.
2. Connect to the server.
3. Enter and send a message
4. Recieve and display the server's reply.
5. Repeat until the user enters bye.
6. Close the connection.

Output:

Client:

Client : Hello
server : Hi
Client : How are you ?
Server : Fine
Client : bye

Server:

Client : Hello
Server : Hi
Client : How are you?
Server : Fine.

Result:

Thus, the Chat client server was implemented using TCP / UDP sockets.

Practical - 13

Aim: To implement our own Ping Program using Python.

Algorithm:

1. Import the os module
2. Get the IP address or hostname from the user.
3. Execute the ping command.
4. Display the result.
5. Stop the program

Program:

```
import os
a = input ("Enter hostname:")
os.system ("ping " + a)
```

Input:

google.com

Output:

Pinging google.com ...
Reply from server

Result:

Thus the ping program was successfully

Practical - 14

Aim:

To implement packet sniffing using RAW sockets in Python.

Algorithm:

1. Import the socket module.
2. Create a RAW socket.
3. Capture network packets
4. Display the packet data
5. Close the socket.

Program:

```
import socket
s = socket.socket(socket.AF_PACKET,
socket.SOCK_RAW, socket.ntohs(3))
while True:
    a = s.recvfrom(65565)
    print(a[0].hex())
```

Input:

Network Packets received by the system

Output:

Displays captured packets in hexadecimal format.

Result: Thus, packet sniffing was implemented successfully using RAW sockets.